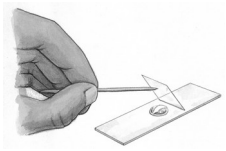
Worksheet 3.3

Preparing a stained wet mount

Answers

Jumbled steps

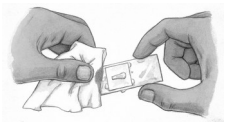
1. The following diagrams show the jumbled steps used to prepare a wet mount of onion epidermis. Place the code letters in the correct sequence.



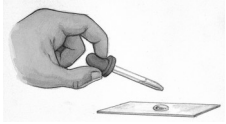
A. Lower a coverslip over the onion skin and water droplet.



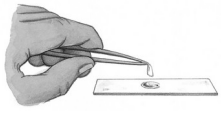
B. Peel a section of onion epidermis.



C. Soak up excess water with a tissue.



D. Add a drop of water to a microscope slide.



E. Place onion skin in drop of water on the slide.

B, D, E, A, C

Correct sequence: .....

Staining cells

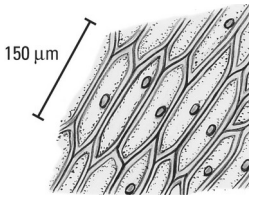
2. It is much easier to see the internal structure of cells by staining the tissue. The following materials are available to prepare a stained wet mount of onion epidermis.

dropping bottle of methylene blue or other similar stain, microscope, microscope slide, coverslip, small section of a peeled onion, forceps, toothpick, dropping bottle of water

(a) Write a method for your investigation.

1. Peel a section of onion epidermis and remove a section of epidermis with forceps.
2. Add a drop of methylene blue to a slide.
3. Place the onion epidermis in the stain.
4. Cover with a coverslip using a toothpick to lower the coverslip.
5. Remove excess stain with a tissue.

(b) The drawing below left was made of the stained onion epidermal tissue.



(i) The nucleus and cell walls take up the methylene blue stain. Use a blue pencil to colour these components in the drawing.

(ii) Use the scale to estimate the greatest width of a cell.

(iii) The experiment was repeated with iodine stain, which strongly stains starch grains a deep blue-black colour. Use coloured pencils to colour the starch grains, which are scattered in the cytoplasm in the cells at the right.



(i) Students colour the cell walls and nuclei blue. (ii) 35–40 µm

(iii) Many tiny starch grains should be coloured blue-black.

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